AMERICAN UNIVERSITY OF BEIRUT
MAROUN SEMAAN FACULTY OF ENGINEERING & ARCHITECTURE

Department of Industrial Engineering and Management (IEM)

Foundation

Engineering Economy

Why are **YOU** taking this course?



What is Engineering Economy?

- Decide where and how to invest capital funds to add value in the future



What is Engineering Economy?

- **Decide** where and how to invest **capital funds** to add **value** in the **future:**
  - Formulate, estimate, and evaluate the expected economic outcomes of alternatives

- Collection of mathematical techniques that simplify economic comparison

- Provides criteria for decision-making



Examples of Decisions?

- **Decide** where and how to invest **capital funds** to add **value** in the **future**

Capital Fund

- Should we spend 30,000$ to **upgrade** equipment to improve productivity?

Future Value

  - Or should we spend 100,000$ and **replace**?
  - Or should we do **nothing**?

Alternatives



Examples of Decisions?

- **Decide** where and how to invest **capital funds** to add **value** in t
- I want to

Your role:
- Design
- Analyze
- Synthesize
- Present

Car				y
Toyota Yaris				
Hybrid Rav4				
Tesla Y	100,000	460hp	-	Medium
Mercedes S-class	300,000	800hp	18L/100KM	Medium

The Decision-making Process

1. Identify and understand the problem; define objective(s)
2. Collect relevant data and information
3. Develop alternatives and estimate costs
4. Define evaluation criteria
5. Evaluate each alternative
6. Select the best alternative
7. Implement the solution
8. Monitor the results
9. Refine the solution

Foundation

Mathematical Modelling and Evaluation

Action and Accountability



Core Concepts



Core Concepts

- Time Value of Money
- Interest
- Economic Equivalence
- Simple and Compound Interest

Time Value of Money

Money Possesses a time value

- If we invest money, we expect to have the original amount + some interest → Money makes money
- If we borrow, we expect to return the original amount + interest

- 1,000$ today is not equivalent to 1,000$ in a year.

Interest

Inflation

Interest

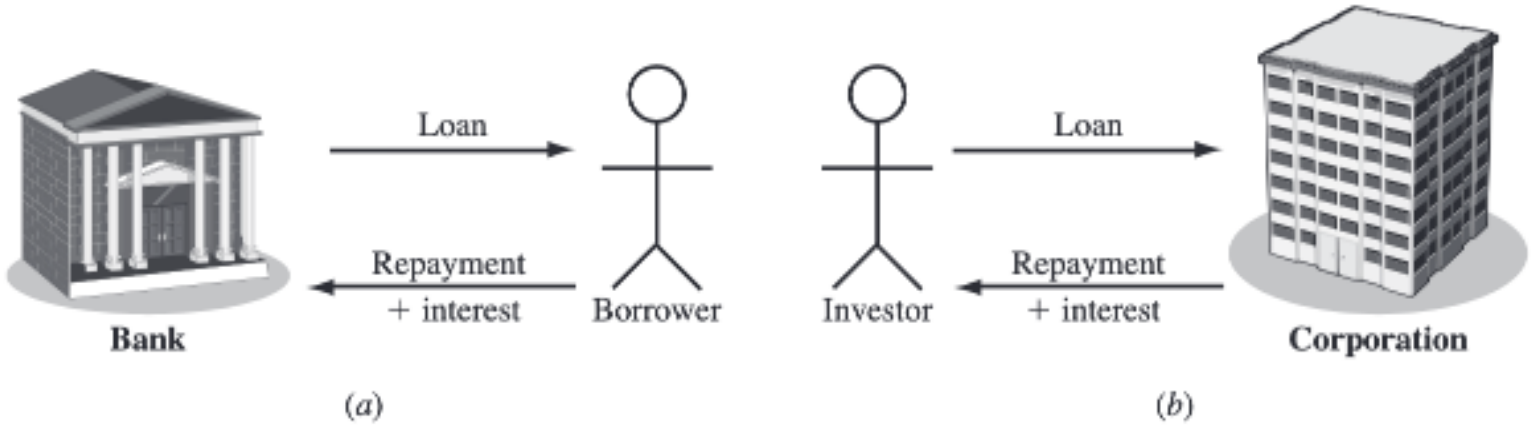


Figure 1–2
(*a*) Interest paid over time to lender. (*b*) Interest earned over time by investor.

Interest – Borrowing Money

- Interest = Final Loan Amount – Principal
- Interest rate:

$$\begin{aligned}\text{interest rate } (\%) &= \frac{\text{interest accrued per time unit}}{\text{principal}} \times 100\% \\ &= \frac{\text{final loan amount} - \text{principal}}{\text{principal}} \times 100\%\end{aligned}$$



Interest – Lending Money

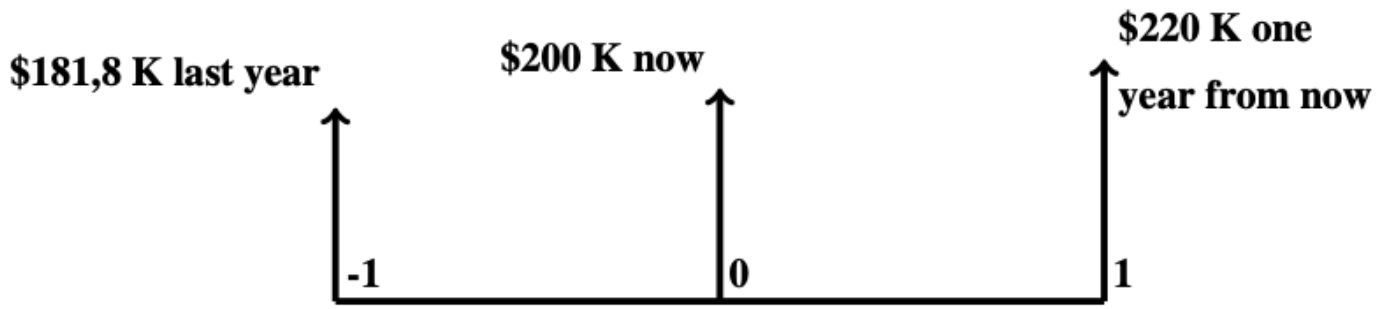
- Interest = Final Investment Value – Principal
- Interest rate (aka rate of return):

$$\text{Rate of return } (\%) = \frac{\text{interest accrued per time unit}}{\text{principal}} \times 100\%$$
$$= \frac{\text{total investment value} - \text{principal}}{\text{principal}} \times 100\%$$



Economic Equivalence

- Knowing the interest rate, we can establish the value of money at any time
  - Assuming an interest rate of 10%:



  - When comparing alternatives, this concepts is important.

$1 today is NOT “equivalent” to $1 a year later.

Economic Equivalence – Example

You want to replace your iPhone. A new one this year is $1100 and it is estimated that the next year's model will be $1200.
At an interest rate of (1) 6%, would you replace it this year or the next one?

- **Present Worth** Analysis

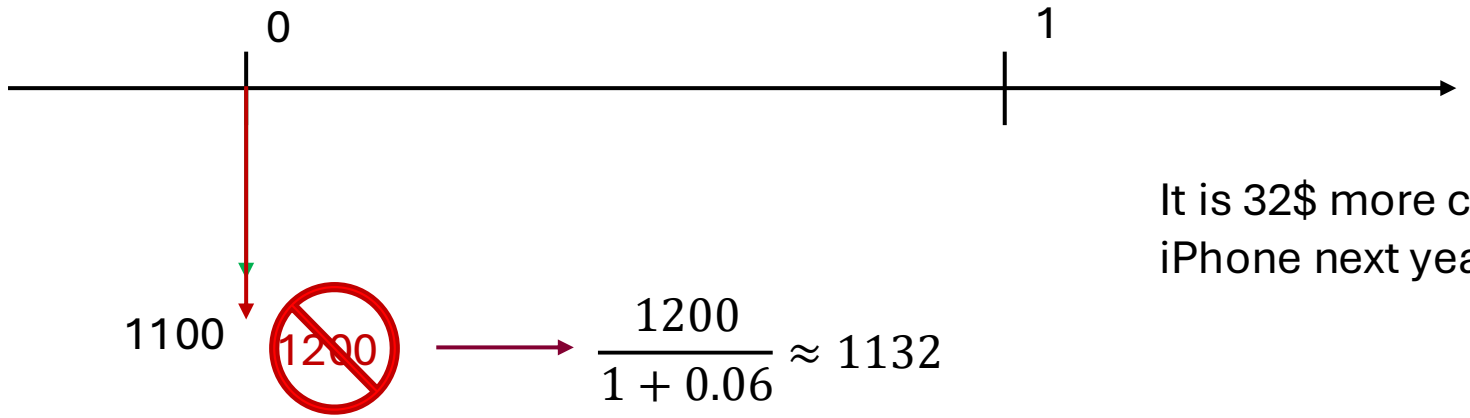


Economic Equivalence – Example

You want to replace your iPhone. A new one this year is $1100 and it is estimated that the next year's model will be $1200.
At an interest rate of (1) 6%, would you replace it this year or the next one?

- **Present Worth** Analysis

$$FW = PW + Interest$$
$$= PW + \text{interest ratio } PW$$
$$= PW(1 + \text{interest ratio})$$
$$PW = \frac{FW}{1 + \text{interest ratio}}$$



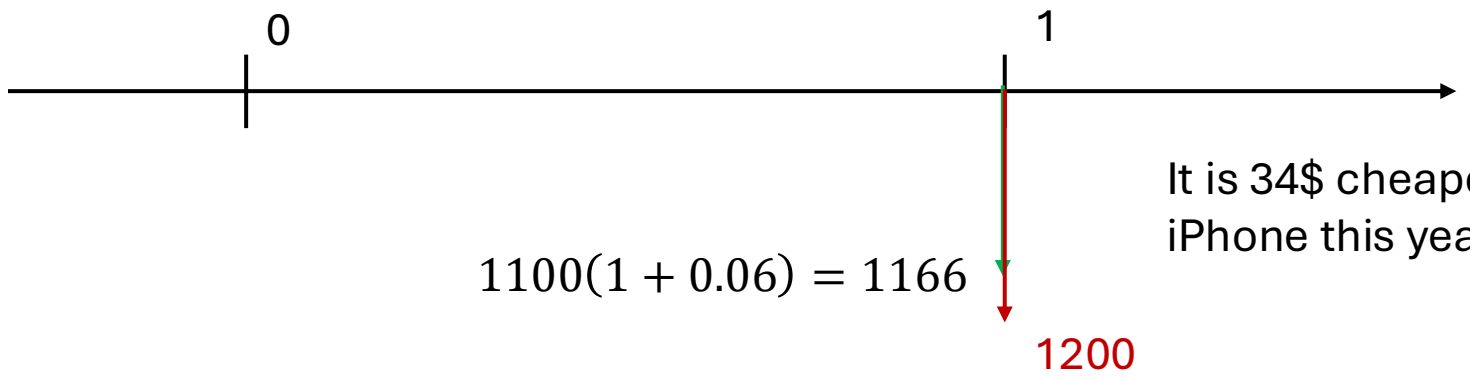
It is 32$ more costly to buy an iPhone next year

Economic Equivalence – Example

You want to replace your iPhone. A new one this year is $1100 and it is estimated that the next year's model will be $1200.
At an interest rate of (1) 6%, would you replace it this year or the next one?

- **Future Worth** Analysis

$$FW = PW + Interest$$
$$= PW + \text{interest ratio } PW$$
$$= PW(1 + \text{interest ratio})$$
$$PW = \frac{FW}{1 + \text{interest ratio}}$$



Interest and Multiple Periods - Simple and Compound

- Interest is paid only on principal

$$
\begin{aligned}
F_n &= P + \underbrace{iP + \cdots + iP}_{n \text{ times}} \\
&= P + niP \\
&= P(1 + ni)
\end{aligned}
$$

- Interest is paid on the principal and interest accrued in previous periods

$$
\begin{aligned}
F_1 &= P + Pi = P(1 + i) \\
F_2 &= F_1 + F_1 i = F_1(1 + i) \\
&= P(1 + i)(1 + i) = P(1 + i)^2 \\
&\vdots \\
F_n &= F_{n-1} + F_{n-1} i = P(1 + i)^n
\end{aligned}
$$

Check This

Simple and Compound Interest – Example

A manufacturing facility plans to upgrade its production line by purchasing an automated packaging machine. The plant manager needs to secure **a loan** of $\$\mathbf{100,000}$ for a duration of $\mathbf{5}$ years and is evaluating two different financing offers (alternatives).

1. A private partner offers the full amount at an annual ***simple interest*** rate of 8.5%.
2. A specialized industrial financing firm offers the loan at an annual ***compound interest*** rate of 6.0%.



Simple and Compound Interest – Example

- Total loan value after 5 years is:

$$\begin{aligned} F_{\text{simple}} &= P(1 + ni_s) \\ &= 100,000(1 + 5 \times 0.085) \\ &= 100,000(1.425) = \$142,500 \end{aligned}$$

- Total loan value after 5 years is:

$$\begin{aligned} F_{\text{compound}} &= P(1 + i_c)^n \\ &= 100,000(1 + 0.06)^5 \\ &= 100,000(1.338226) \\ &= \$133,822.56 \end{aligned}$$

Cash Flows

The most important tool you will learn



Cash Flows

Cash flows are the amounts of money estimated for future projects or observed for project **events** that have taken place

- Cash Inflow
- Cash outflow
- **Cashflow Diagram**



Cash Inflow

- Amount of funds flowing into a firm.

Receipts, revenues, incomes, savings

  - Sales Revenue: +$100,000 from selling an equipment
  - Borrowed Money: +$50,000 to install a solar system
  - Savings: +$2,000 per month on electricity cost
  - Asset salvage value: +$15,000 from selling the CNC machine at the end of its 5-year useful life
  - Income tax savings: +$4,500 annually due to equipment depreciation deductions

Cash Outflow

- Amount of funds flowing out of a firm.

costs, disbursements, expenses, and taxes

- Equipment purchase: -$50,000 to acquire an energy management system
- Installation costs: -$5,000 for site preparation and system integration
- Operating and maintenance: -$1,500 per month for tooling, lubrication, and routine servicing
- Labor expenses: -$4,000 - $8000 per month for hiring a specialized machine operator
- Loan repayment: -$22,000 annually for principal and interest payments to the industrial bank
- Income tax liability: -$8,000 annually due to increased taxable profits from higher production capacity



Cashflow Diagram*

- Timeline
- **End-of-period** cash flow
- The diagram depends on the point of view (investor vs. company)
- Cash inflows are upward arrows
- Cash outflows are downward arrows







Cashflow Diagram* - Example

- Jeff took a $10,000 loan at a rate of 5% over 5 periods to buy anew hybrid car.
- He also needs to pay insurance of $800 a year
- It is expected that he would save $200 monthly in fuel consumption
- However, maintenance is more expensive by around $200 per year.

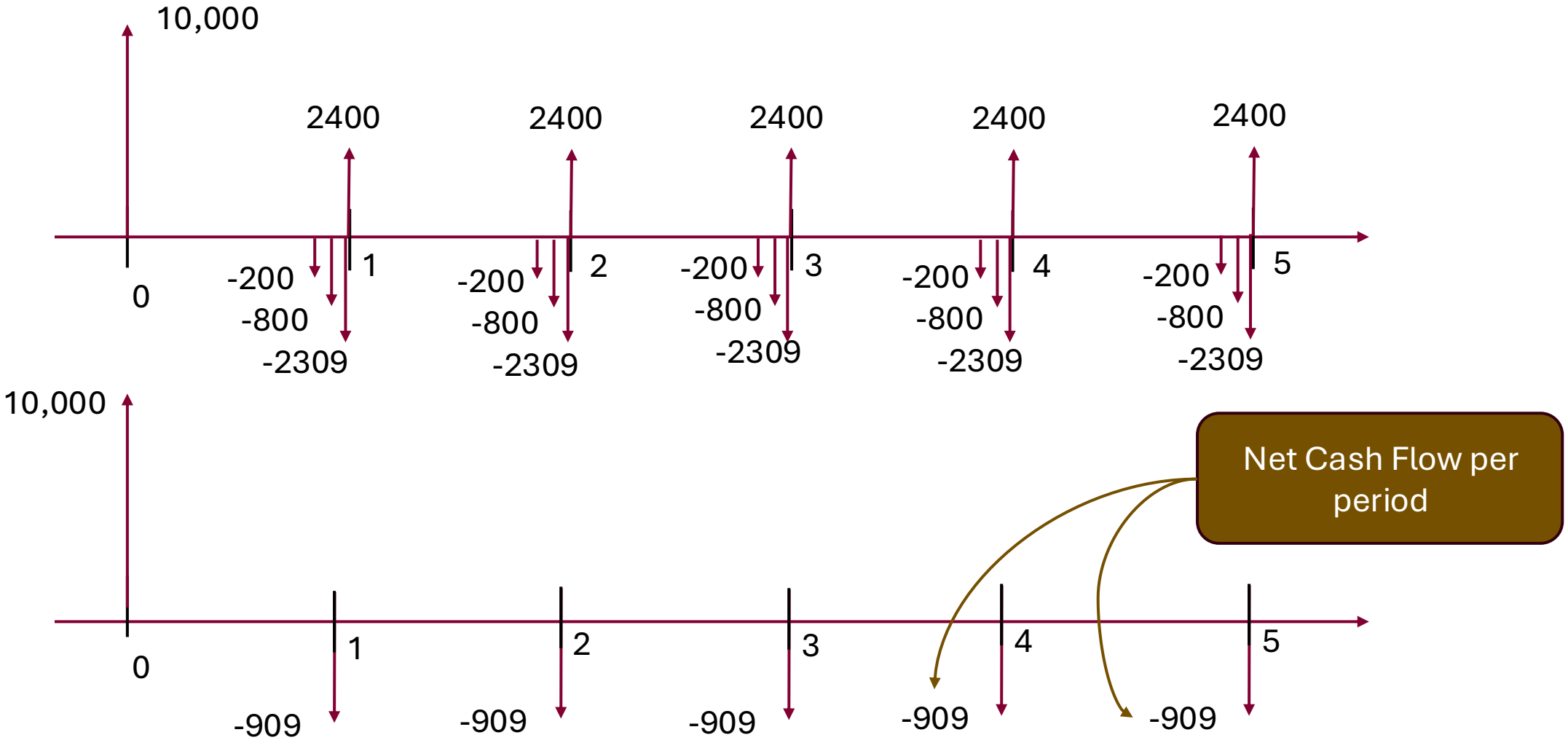
Cashflow Diagram* - Example

- Jeff took a $10,000 loan at a rate of 5% over 5 periods to buy a new hybrid car.→ Yearly loan payment of 2309*
- He also needs to pay insurance of $800 a year
- It is expected that he would save $200 monthly in fuel consumption → Yearly savings of 2400
- However, maintenance is more expensive by around $200 per year.



* We will see how to get this value in detail

Cashflow Diagram* - Example



Cash Flow – Example 2

- In a company's renovation of a small office building, two feasible alternatives for upgrading the HVAC system have been identified.
- Alternative A: rebuild (overhaul) the existing HVAC system
- Alternative B: install a new HVAC system that utilizes existing ductwork must be implemented.
- At the end of eight years, the estimated market value for Alternative A is $2,000$ and for Alternative B it is $8,000$

	Alternative A	Alternative B
Equipment, labor, and materials	18,000	60,000
Annual cost of electricity	32,000	9,000
Annual maintenance expenses	2,400	16,000
Replacement of a major component four years	-	9,400

❑Use a cash-flow table and end-of-year convention to tabulate the net cash flows for both alternatives.

❑Determine the annual net cash-flow difference between the alternatives (B - A).



Minimum Attractive Rate of Return

MARR



Minimum Attractive Rate of Return (MARR)

- An investment is **profitable** if the **rate of return** is positive:

$$\text{Rate of Return} = \frac{\text{Interest accrued per time unit}}{\text{Principal}} 100\%$$

- If you invest $1000, after a year it becomes $1010.
  - Is this a good investment? And why?



Minimum Attractive Rate of Return (MARR)

- To be considered financially viable, a project's **expected** ROR **must** meet or exceed the $MARR$.

$$ROR \geq MARR$$

- MARR:
  - is a reasonable rate of return established for the evaluation and selection of alternatives (including the do nothing)
  - established by company management
  - is estimated based on the weighted average cost of capital of sources of funding available to the firm.



Minimum Attractive Rate of Return (MARR)

- Established by company management, considering:
  - Cost and Source of Capital: The amount of money available for investment, and whether those funds are equity or debt.
    - Equity financing: the firm uses its own assets to finance often through issuing stocks.
    - Debt financing: the firm borrows money to finance often through issuing bonds.
  - Project Portfolio: Number of projects, urgency, essential or elective
  - Project Risk: Riskier projects require a higher MARR
  - Organization Type



Minimum Attractive Rate of Return (MARR)

- Estimated based on the **weighted average cost of capital (WACC)** of sources of funding available to the firm.
- You can buy the TV with:
  - 40% credit card money at 15% per year
  - and 60% from the savings funds that is earning you 5% per year.

$$\begin{aligned}\text{WACC} &= (0.40 \times 15\%) + (0.60 \times 5\%) \\ &= 6\% + 3\% = 9\% \text{ per year}\end{aligned}$$

$$\text{ROR} \geq \text{MARR} > \text{Cost of Capital}$$

Rule of 72

- This rule (approximately) estimates the number of time periods, *n*, it takes for an amount of money to double under an ROR of *i* (%):

$$n = \frac{72}{i}$$

- Can a $1 million project double its value in 5 years, if the project expected ROR=10%?
- Translate an MARR of 12% into something tangible!!



Key Takeaways

- Systematic way for comparing and choosing alternatives
- Money possesses time value
  - Interest
  - Economic equivalence
- Cash flows
  - in- and outflows
  - Net cash flow
  - Diagrams and tables
- MARR

